

5. Tensor fields

5.1. A crash course on vector bundles. Let X be a real manifold. A **vector bundle** on X is, informally speaking, a (locally trivial) fiber bundle on X whose fibers are finite dimensional vector spaces. In other words, it is a family of vector spaces parametrized by $x \in X$ and varying regularly with x . More precisely, we have the following definition.

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

Definition 5.1. A $\mathbb{K}$ -**vector bundle** of rank n on X is a manifold E with a surjective regular map $p : E \rightarrow X$ and a $\mathbb{K}$ -vector space structure on each fiber $p^{-1}(x)$ such that every $x \in X$ has a neighborhood U admitting a diffeomorphism $g : U \times \mathbb{K}^n \rightarrow p^{-1}(U)$ with the following properties:

- (i) $(p \circ g)(u, v) = u$, and
- (ii) the map g is $\mathbb{K}$ -linear on the second factor.

In other words, locally on X , E is isomorphic to $X \times \mathbb{K}^n$, but not necessarily globally so.

As for ordinary fiber bundles, E is called the **total space** and X the **base** of the bundle.

Note that even if X is a complex manifold and $\mathbb{K} = \mathbb{C}$, E need not be a complex manifold.

Definition 5.2. A complex vector bundle $p : E \rightarrow X$ on a complex manifold X is said to be **holomorphic** if E is a complex manifold and the diffeomorphisms g_U can be chosen holomorphic.

From now on, unless specified otherwise, all complex vector bundles on complex manifolds we consider will be holomorphic.

It follows from the definition that if $p : E \rightarrow X$ is a vector bundle then X has an open cover $\{U_\alpha\}$ such that E trivializes on each U_α , i.e., there is a diffeomorphism $g_\alpha : U_\alpha \times \mathbb{K}^n \rightarrow p^{-1}(U_\alpha)$ as above. In this case we have **clutching functions**

$$h_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL_n(\mathbb{K})$$

(holomorphic if E is a holomorphic bundle), defined by the formula

$$(g_\alpha^{-1} \circ g_\beta)(x, v) = (x, h_{\alpha\beta}(x)v)$$

which satisfy the **consistency conditions**

$$h_{\alpha\beta}(x) = h_{\beta\alpha}(x)^{-1}$$

and

$$h_{\alpha\beta}(x) \circ h_{\beta\gamma}(x) = h_{\alpha\gamma}(x)$$

for $x \in U_\alpha \cap U_\beta \cap U_\gamma$. Moreover, the bundle can be reconstructed from this data, starting from the disjoint union $\sqcup_\alpha U_\alpha \times \mathbb{K}^n$ and identifying (gluing) points according to

$$h_{\alpha\beta} : (x, v) \in U_\beta \times \mathbb{K}^n \sim (x, h_{\alpha\beta}(x)v) \in U_\alpha \times \mathbb{K}^n.$$

The consistency conditions ensure that the relation $\sim$ is symmetric and transitive, so it is an equivalence relation, and we define E to be the space of equivalence classes with the quotient topology. Then E has a natural structure of a vector bundle on X .

This can also be used for constructing vector bundles. Namely, the above construction defines a $\mathbb{K}$ -vector bundle on X once we are given a cover $\{U_\alpha\}$ on X and a collection of clutching functions

$$h_{\alpha\beta} : U_\alpha \cap U_\beta \rightarrow GL_n(\mathbb{K})$$

satisfying the consistency conditions.

Remark 5.3. All this works more generally for non-linear fiber bundles if we drop the linearity conditions along fibers.

Example 5.4. 1. The **trivial bundle** $p : E = X \times \mathbb{K}^n \rightarrow X$, $p(x, v) = x$.

2. The **tangent bundle** is the vector bundle $p : TX \rightarrow X$ constructed as follows. For the open cover we take an atlas of charts (U_α, ϕ_α) with transition maps

$$\theta_{\alpha\beta} = \phi_\alpha \circ \phi_\beta^{-1} : \phi_\beta(U_\alpha \cap U_\beta) \rightarrow \phi_\alpha(U_\alpha \cap U_\beta),$$

and we set

$$h_{\alpha\beta}(x) := d_{\phi_\beta(x)}\theta_{\alpha\beta}.$$

(Check that these maps satisfy consistency conditions!)

Thus the tangent bundle TX is a vector bundle of rank $\dim X$ whose fiber $p^{-1}(x)$ is naturally the tangent space $T_x X$ (indeed, the tangent vectors transform under coordinate changes exactly by multiplication by $h_{\alpha\beta}(x)$). In other words, it formalizes the idea of “the tangent space $T_x X$ varying smoothly with $x \in X$ ”.

Definition 5.5. A **section** of a map $p : E \rightarrow X$ is a map $s : X \rightarrow E$ such that $p \circ s = \text{Id}_X$.

Example 5.6. If $p : X \times Y = E \rightarrow X$, $p(x, y) = x$ is the trivial bundle then a section $s : X \rightarrow E$ is given by $s(x) = (x, f(x))$ where $y = f(x)$ is a function $X \rightarrow Y$, and the image of s is the graph of f . So the notion of a section is a generalization of the notion of a function.

In particular, we may consider sections of a vector bundle $p : E \rightarrow X$ over an open set $U \subset X$. These sections form a vector space denoted $\Gamma(U, E)$.

Exercise 5.7. Show that a vector bundle $p : E \rightarrow X$ is trivial (i.e., globally isomorphic to $X \times \mathbb{K}^n$) if and only if it admits sections $s_1, \dots, s_n$ which form a basis in every fiber $p^{-1}(x)$.

5.2. Vector fields.

Definition 5.8. A **vector field** on X is a section of the tangent bundle TX .

Thus in local coordinates a vector field looks like

$$\mathbf{v} = \sum_i v_i \frac{\partial}{\partial x_i},$$

$v_i = v_i(\mathbf{x})$, and if $x_i \mapsto x'_i$ is a change of local coordinates then the expression for $\mathbf{v}$ in the new coordinates is

$$\mathbf{v} = \sum_i v'_i \frac{\partial}{\partial x'_i}$$

where

$$v'_i = \sum_j \frac{\partial x'_i}{\partial x_j} v_j,$$

i.e., the clutching function is the **Jacobi matrix** of the change of variable. Thus, every vector field $\mathbf{v}$ on X defines a derivation of the algebra $O(U)$ for every open set $U \subset X$ compatible with restriction maps $O(U) \rightarrow O(V)$ for $V \subset U$;⁵ in particular, a derivation $O_x \rightarrow O_x$ for all $x \in X$. Conversely, it is easy to see that such a collection of derivations gives rise to a vector field, so this is really the same thing.

A manifold X is called **parallelizable** if its tangent bundle is trivial. By Exercise 5.7, this is equivalent to having a collection of vector fields $\mathbf{v}_1, \dots, \mathbf{v}_n$ which form a basis in every tangent space (such a collection is called a **frame**). For example, the circle S^1 and hence the torus $S^1 \times S^1$ are parallelizable. On the other hand, the sphere S^2 is not parallelizable, since it does not even have a single nowhere vanishing vector field (the **Hairy Ball theorem**, or **Hedgehog theorem**). The same is true for any even-dimensional sphere S^{2m} , $m \geq 1$.

⁵In other words, using a fancier language, $\mathbf{v}$ defines a derivation of the **sheaf** of regular functions on X .

5.3. Tensor fields, differential forms. Since vector bundles are basically just smooth families of vector spaces varying over some base manifold X , we can do with them the same things we can do with vector spaces - duals, tensor products, symmetric and exterior powers, etc. E.g., the **cotangent bundle** T^*X is dual to the tangent bundle TX .

More generally, we make the following definition.

Definition 5.9. A **tensor field** of rank (k, m) on a manifold X is a section of the tensor product $(TX)^{\otimes k} \otimes (T^*X)^{\otimes m}$.

For example, a tensor field of rank $(1, 0)$ is a vector field. Also, a skew-symmetric tensor field of rank $(0, m)$ is called a **differential m -form** on X . In other words, a differential m -form is a section of the vector bundle $\Lambda^m T^*X$.

For instance, if $f \in O(X)$ then we have a differential 1-form df on X , called **the differential of f** (indeed, recall that $d_x f : T_x X \rightarrow \mathbb{K}$). A general 1-form can therefore be written in local coordinates as

$$\omega = \sum_i a_i dx_i.$$

where $a_i = a_i(\mathbf{x})$. If coordinates are changed as $x_i \mapsto x'_i$, then in new coordinates

$$\omega = \sum_i a'_i dx'_i$$

where

$$a'_i = \sum_j \frac{\partial x_j}{\partial x'_i} a_j.$$

Thus the clutching function is the **inverse of the Jacobi matrix** of the change of variable. For instance,

$$df = \sum_i \frac{\partial f}{\partial x_i} dx_i.$$

More generally, a differential m -form in local coordinates looks like

$$\omega = \sum_{1 \leq i_1 < \dots < i_m \leq n} a_{i_1 \dots i_m}(x) dx_{i_1} \wedge \dots \wedge dx_{i_m}.$$

5.4. Left and right invariant tensor fields on Lie groups. Note that if a Lie group G acts on a manifold X , then it automatically acts on the tangent bundle TX and thus on vector and, more generally, tensor fields on X . In particular, G acts on tensor fields on itself by left and right translations; we will denote this action by L_g and R_g ,

respectively. We say that a tensor field T on G is **left invariant** if $L_g T = T$ for all $g \in G$, and **right invariant** if $R_g T = T$ for all $g \in G$.

Proposition 5.10. (i) For any $\tau \in \mathfrak{g}^{\otimes k} \otimes \mathfrak{g}^{*\otimes m}$ there exists a unique left invariant tensor field $\mathbf{L}_\tau$ and a unique right invariant tensor field $\mathbf{R}_\tau$ whose value at 1 is τ . Thus, the spaces of such tensor fields are naturally isomorphic to $\mathfrak{g}^{\otimes k} \otimes \mathfrak{g}^{*\otimes m}$.

(ii) $\mathbf{L}_\tau$ is also right invariant iff $\mathbf{R}_\tau$ is also left invariant iff τ is invariant under the adjoint representation Ad_g .

Proof. We only prove (i). Consider the tensor fields $\mathbf{L}_\tau(g) := L_g \tau$, $\mathbf{R}_\tau(g) := R_{g^{-1}} \tau$ (i.e., we “spread” τ from $1 \in G$ to other points $g \in G$ by left/right translations). By construction, $R_{g^{-1}} \tau$ is right invariant, while $L_g \tau$ is left invariant, both with value τ at 1, and it is clear that these are unique. $\square$

Exercise 5.11. Prove Proposition 5.10(ii).

Corollary 5.12. A Lie group is parallelizable.

Proof. Given a basis $e_1, \dots, e_n$ of $\mathfrak{g} = T_1 G$, the vector fields $L_g e_1, \dots, L_g e_n$ form a frame. $\square$

Remark 5.13. In particular, S^1 and $SU(2) = S^3$ are parallelizable. It turns out that S^n for $n \geq 1$ is parallelizable if and only if $n = 1, 3, 7$ (a deep theorem in differential topology). So spheres of other dimensions don’t admit a Lie group structure. The sphere S^7 does not admit one either, although it admits a weaker structure of a “homotopy Lie group”, or H -space (arising from octonions) which suffices for parallelizability. Thus the only spheres admitting a Lie group structure are $S^0 = \{1, -1\}$, S^1 and S^3 . This result is fairly elementary and will be proved in Section 46.

MIT OpenCourseWare
<https://ocw.mit.edu>

18.745 Lie Groups and Lie Algebras I

Fall 2020

For information about citing these materials or our Terms of Use, visit: <https://ocw.mit.edu/terms>.